**CABINET MÉDICAL DES TILLEULS**

**CMT VALLON**

14 allée des Tilleuls

71990 Vallon-sur-Brume

Tél: 03 85 10 20 30

Réf. cabinet : 2026-0417

Docteur Claire FONTAINE

Ophtalmologiste

Ancienne assistante des hôpitaux de Lyon

Diplômée de l'université de Bourgogne

MALADIES ET CHIRURGIE DES YEUX

RPPS : 10000000417

Vallon le 11/05/2026

Monsieur Paul MARCHAND

# ORDONNANCE DE LUNETTES

Une paire de lunettes avec monture

Simple foyer

**Oeil droit: -2.75 (-0.25) 170°**

**Oeil gauche: -2.75 (-0.50) 165°**

Signature et cachet

# Invoice

Elena Martin | ID: DOC-101-1 | 2026-10-15

From: Northfield Studio

Bill to: Elena Martin

| Description | Qty | Unit price | Amount |
|---|---|---|---|
| Office paper | 1 | 12.35 | 12.35 |
| Delivery service | 2 | 15.55 | 31.10 |
| Maintenance kit | 3 | 18.75 | 56.25 |
| Consultation | 4 | 21.95 | 87.80 |
| Storage boxes | 5 | 25.15 | 125.75 |
| Printer ink | 6 | 28.35 | 170.10 |

## Total due

739.55

IBAN: FR76 3000 6000 0112 3456 7890 189

billing@example.test

# 文档转换质量研究

陈晓明 | ID: DOC-109-3 | 2026-10-15

## 1 引言

服务从2026年10月15日开始，持续十二个月。开展额外工作之前必须获得书面批准。每笔付款都必须注明参考编号。双方保留原始文件，并在收到通知之后确认变更。

服务从2026年10月15日开始，持续十二个月。开展额外工作之前必须获得书面批准。每笔付款都必须注明参考编号。双方保留原始文件，并在收到通知之后确认变更。

服务从2026年10月15日开始，持续十二个月。开展额外工作之前必须获得书面批准。每笔付款都必须注明参考编号。双方保留原始文件，并在收到通知之后确认变更。

服务从2026年10月15日开始，持续十二个月。开展额外工作之前必须获得书面批准。每笔付款都必须注明参考编号。双方保留原始文件，并在收到通知之后确认变更。

服务从2026年10月15日开始，持续十二个月。开展额外工作之前必须获得书面批准。每笔付款都必须注明参考编号。双方保留原始文件，并在收到通知之后确认变更。

## 2 结果

F1 = 2PR / (P + R); n = 128; p = 0.032.

| 项目名称 | 数量 | 单价 | 金额 |
|---|---|---|---|
| 办公用纸 | 1 | 12.35 | 12.35 |
| 配送服务 | 2 | 15.55 | 31.10 |
| 维护套件 | 3 | 18.75 | 56.25 |

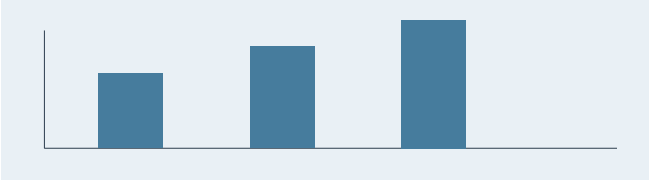


图1 季度观测结果

## 参考文献

[1] Martin, E. Document analysis. 2024. pp. 12-18.

# Physics 101 — Mid-term Examination

Time: 60 min · Total: 40 marks

Name: _____ Date: _____ Class: _____

**Section A — Multiple choice.** Circle one answer.

**1.** A car accelerates uniformly from rest to 24 m/s in 8.0 s. Its acceleration is: *[2]*

(a) 1.5 m/s$^2$ (b) 3.0 m/s$^2$

(c) 6.0 m/s$^2$ (d) 192 m/s$^2$

**2.** The kinetic energy of a 2.0 kg object moving at 5.0 m/s is given by $E = \tfrac{1}{2}mv^2$ = *[2]*

(a) 10 J (b) 20 J

(c) 25 J (d) 50 J

**Section B — Short answer.**

**3.** State Newton's second law and write it as an equation. *[3]*

_____

_____

**4.** A 0.50 kg ball is thrown vertically upward at 12 m/s. Calculate (a) the maximum height reached and (b) the time to return to the thrower. Take $g = 9.8$ m/s$^2$. *[6]*

_____

_____

_____

**5.** Explain, with one example, why a net force of zero does not imply that an object is at rest. *[3]*

_____

_____

# Certificate of Achievement

This certificate is proudly presented to

## *Amara Nwosu-Fischer*

for outstanding completion of the Advanced Data Stewardship Programme
with distinction, awarded on the fourth of November, two thousand twenty-six.

Certificate No. ADS-2026-00417 · 120 contact hours · Grade 94 / 100

Dr. Helena Brandt
Programme Director

Prof. Tomas Ortega
Dean of Studies

OFFICIAL
SEAL 2026

**SCHEDULE B (Form 1040)**

Department of the Treasury
Internal Revenue Service

# Interest and Ordinary Dividends

**Attach to Form 1040 or 1040-SR.**
**Go to *www.irs.gov/ScheduleB* for instructions and the latest information.**

OMB No. 1545-0074

**2025**

Attachment Sequence No. **08**

Name(s) shown on return

**Your social security number**

## Part I Interest

(See instructions and the Instructions for Form 1040, line 2b.)

**Note:** If you received a Form 1099-INT, Form 1099-OID, or substitute statement from a brokerage firm, list the firm's name as the payer and enter the total interest shown on that form.

| | | | Amount |
|---|---|---|---|
| 1 | List name of payer. If any interest is from a seller-financed mortgage and the buyer used the property as a personal residence, see the instructions and list this interest first. Also, show that buyer's social security number and address: | 1 | |
| 2 | Add the amounts on line 1 | 2 | |
| 3 | Excludable interest on series EE and I U.S. savings bonds issued after 1989. Attach Form 8815 | 3 | |
| 4 | Subtract line 3 from line 2. Enter the result here and on Form 1040 or 1040-SR, line 2b | 4 | |

**Note:** If line 4 is over $1,500, you must complete Part III.

## Part II Ordinary Dividends

(See instructions and the Instructions for Form 1040, line 3b.)

**Note:** If you received a Form 1099-DIV or substitute statement from a brokerage firm, list the firm's name as the payer and enter the ordinary dividends shown on that form.

| | | | Amount |
|---|---|---|---|
| 5 | List name of payer: | 5 | |
| 6 | Add the amounts on line 5. Enter the total here and on Form 1040 or 1040-SR, line 3b | 6 | |

**Note:** If line 6 is over $1,500, you must complete Part III.

## Part III Foreign Accounts and Trusts

You must complete this part if you (**a**) had over $1,500 of taxable interest or ordinary dividends; (**b**) had a foreign account; or (**c**) received a distribution from, or were a grantor of, or a transferor to, a foreign trust.

**Caution:** If required, failure to file FinCEN Form 114 may result in substantial penalties. Additionally, you may be required to file Form 8938, Statement of Specified Foreign Financial Assets. See instructions.

| | | Yes | No |
|---|---|---|---|
| 7a | At any time during 2025, did you have a financial interest in or signature authority over a financial account (such as a bank account, securities account, or brokerage account) located in a foreign country? See instructions | | |
| | If "Yes," are you required to file FinCEN Form 114, Report of Foreign Bank and Financial Accounts (FBAR), to report that financial interest or signature authority? See FinCEN Form 114 and its instructions for filing requirements and exceptions to those requirements | | |
| b | If you are required to file FinCEN Form 114, list the name(s) of the foreign country(-ies) where the financial account(s) is (are) located: | | |
| 8 | During 2025, did you receive a distribution from, or were you the grantor of, or transferor to, a foreign trust? If "Yes," you may have to file Form 3520. See instructions | | |

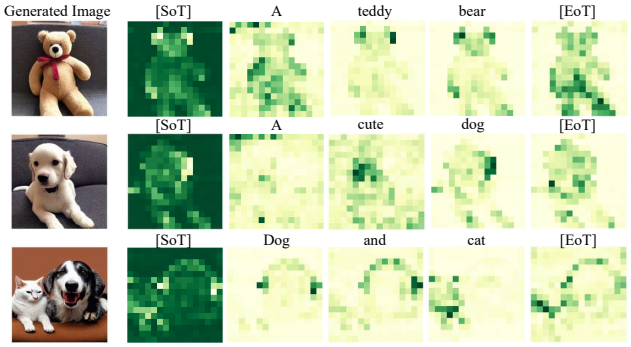



Figure 3: Cross-attention maps of different text prompts at the generation process. All of the examples indicate that start token ( [SoT] ) and padding tokens ( [EoT] ) carry rich semantic and layout information.

There is one more aspect of the SD architecture design that we require for our purposes. While there are several important design choices that make the network $D$ work well in practice, the mechanism that is of interest in our investigation is *cross-attention*, which allows the generation process to be conditioned on text, thus connecting visual and textual information. Each cross-attention layer takes an intermediate feature tensor $\boldsymbol{z}^{(\gamma)} \in \mathbb{R}^{C \times \frac{H}{r} \times \frac{W}{r}}$ as input, where $\gamma$ is the index of the relevant layer in the network, and $r$ is a scaling factor defining the spatial resolution at that level of the representation. The cross-attention map $A^{(\gamma)}$ associates each spatial location $u \in \{1, \dots, \frac{H}{r}\} \times \{1, \dots, \frac{W}{r}\}$ to a token indexed by $i \in \{1, \dots, N\}$:

$$A_{ui}^{(\gamma)} = \frac{\exp\langle \boldsymbol{z}_u^{(\gamma)}, K_i^{(\gamma)} \rangle}{\sum_{j=1}^{N} \exp\langle \boldsymbol{z}_u^{(\gamma)}, K_j^{(\gamma)} \rangle}, \quad \boldsymbol{a}_u^{(\gamma)} = \sum_{i=1}^{N} A_{ui}^{(\gamma)} V_i^{(\gamma)},$$

where the value $V_i^{(\gamma)}$ and the key $K_i^{(\gamma)}$ are linear transformations of the token embedding $Y_i$ provided by the textual encoder, and $\boldsymbol{a}_u^{(\gamma)}$ is the output of the cross-attention layer.

## 3.2. Layout Guidance

Next, we introduce our method for training-free layout guidance. It has already been shown that cross-attention layers regulate the spatial layout of the generated image [12]. Specifically, the attention map the attention map $A_{ui}^{(\gamma)}$ determines how strongly each location $u$ in layer $\gamma$ is associated with each of the $N$ text tokens $y_i$. Since the sum of association strengths $\sum_{i=1}^{N} A_{ui}^{(\gamma)} = 1$ for each spatial location $u$, the different tokens can be thought of as "competing" for association with that location. To control the image layout using a bounding box B that corresponds to token $y_i$, the attention can be biased such that locations $u \in B$ within the target box are strongly associated with the token $y_i$ (while other locations are not). We introduce two ways of

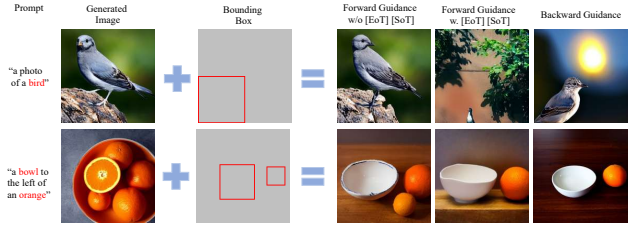



Figure 4: Comparison between forward and backward guidance, including guidance of start and padding tokens.

achieving this effect, which we refer to as forward guidance (Section 3.2.1) and backward guidance (Section 3.2.2). We give an overview of the two strategies in Figure 2.

### 3.2.1 Forward Guidance

In forward guidance, we represent the bounding box $B$ as a smooth windowing function $g_u^{(\gamma)}$ which is equal to a constant $c > 0$ inside the box and quickly falls to zero outside.[1] We rescale the windowing function to sum to one ($\|g^{(\gamma)}\|_1 = 1$). Then, we bias the attention map $A^{(\gamma)}$ by replacing it via:

$$A_{ui}^{(\gamma)} \leftarrow (1 - \lambda) A_{ui}^{(\gamma)} + \lambda g_u^{(\gamma)} \sum_v A_{vi}^{(\gamma)}, \tag{1}$$

where $\lambda \in [0, 1]$ defines the strength of the intervention. Note that: (1) only the cross-attention map $A_{ui}^{(\gamma)}$ corresponding to token $y_i$ is affected by this intervention (2) the window is weighed by the mass $\sum_v A_{vi}^{(\gamma)}$ so as to leave the latter unchanged.

This intervention is injected at each application of the denoiser function $D$ at a selected subset of layers $\gamma \in \Gamma$. This means that the activation maps computed by each of the selected layers are replaced with Equation (1).

### 3.2.2 Backward Guidance

The second approach we consider is backward guidance. In this case, we bias the attention $A_{ui}^{(\gamma)}$ by introducing the energy function

$$E(A^{(\gamma)}, B, i) = \left(1 - \frac{\sum_{u \in B} A_{ui}^{(\gamma)}}{\sum_u A_{ui}^{(\gamma)}}\right)^2 \tag{2}$$

This is used in a manner similar to *classifier guidance* [3]. Specifically, at each application of the denoiser $D$, when layer $\gamma \in \Gamma$ is evaluated, the gradient of the loss (2) is computed via backpropagation to update the latent $\boldsymbol{z}_t (\equiv \boldsymbol{z}_t^{(0)})$:

$$\boldsymbol{z}_t \leftarrow \boldsymbol{z}_t - \sigma_t \eta \nabla_{\boldsymbol{z}_t} \sum_{\gamma \in \Gamma} E(A^{(\gamma)}, B, i), \tag{3}$$

[1] For simplicity, in our implementation, we put a Gaussian blob with $\sigma$ decided by the resolution, height, and width of the bounding box.